

Title of the project

AI-Based Live Interview and Automated Onboarding System

Student Name

NAME1:ANSARI AZKA SHAKEEL AHMED

NAME2:KHAN ARSHIYA BANO FERROZ

Internal Guide

Prof.Agah Safa Mam

Project Synopsis

1. Title of the Project

AI-Based Live Interview and Automated Onboarding System

2. Introduction

This project is basically made for reduce HR work and Use AI technology to make HR work easy and reducing time. In this project we are make AI panel to conduct interview based on employee skill and experience that generate questions. Its live streaming interview to record also employee interviews answer and based on employee answers AI given the result. The after first round pass via managers conducted interview. Here is main function performing AI we can say mini AI-HR they automatically save the users data in database and automatically send the email to HR and admin/manager panel.

This system aims to improve recruitment efficiency, reduce manual workload, and ensure structured and unbiased evaluation.

3. Objectives/ Existing System and Need for System

Existing system problems:

- to reduce manual work
- time saving
- high HR work
- risk for unfair practice

Need for system:

- To automate the technical interview using AI
- Live-video based AI-assisted interview
- Dynamically generate question based on candidate profile
- To generate structured interview reports for HR
- Less chance to cheating due to monitor behavior using anti-cheating detection

4. Scope of the Project

- Live AI video for interview
- Speech to text conversion
- Interview recording storage
- Based on employee profile generate questions
- Main function AI technology use

Limitations:

- Internet connection required for live streaming
- Final hiring decision remains with HR

5. Technologies Used

◆ Programming Language:

- Python

◆ Backend Framework:

- FastAPI / Flask

◆ Frontend:

- HTML
- CSS
- JavaScript
- WebRTC (for live video streaming)

◆ AI & NLP:

- OpenAI API / LLM
- Whisper (Speech-to-Text)
- TF-IDF / Keyword-based scoring

◆ Database:

- PostgreSQL / MySQL / SQLite

◆ Other Tools:

- OpenCV (for basic face detection)
- MediaPipe (for gaze detection – optional)
- Git & GitHub

6. System Architecture

◆ High-Level Architecture Description:

The system follows a client-server architecture.

1. The **Candidate Interface (Frontend)** captures user profile details and live video stream.
2. The **Backend Server (Python-based API)** manages interview logic, question generation, evaluation, and storage.
3. The **AI Engine** processes speech-to-text conversion and answer evaluation.
4. The **Database** stores candidate data, interview records, and scores.
5. The **HR Dashboard** displays reports and recommendations.

7. Modules and Functionalities

Module 1: Live AI-Interview Module

- AI Generate Question according to profile
- Interview recording
- WebRTC-based live video streaming

Module 2: Candidate Management Module

- Candidate login and register
- Automate profile data collection(skills, qualification, experience)
- Upload CV

Module 3: AI reporting & Behavioral Monitoring Module

- AI based generate final score
- Face detection
- Multiple person detection
- HR review dashboard

8. Methodology

The project follows the **Agile Development Model**.

Phases:

1. Requirement Analysis
2. System Design
3. Module Development
4. Testing and Debugging
5. Integration
6. Deployment

Agile methodology allows continuous improvement and iterative development of features.

9. Expected Outcome

- HR workload reduce
- Time reducing
- Automate the question generated
- Faster requirement process
- AI assisted lie interview
- **The system will demonstrate how Artificial Intelligence can enhance modern recruitment processes**

10. Conclusion

The AI-Based Live Interview and Automated Onboarding System aims to modernize traditional recruitment by integrating AI technologies such as NLP, speech recognition, and adaptive questioning. The system reduces manual effort, ensures structured evaluation, and provides intelligent decision support to HR teams.

This project highlights the practical application of Artificial Intelligence in real-world recruitment and onboarding processes.

11. References *(if any)*

- FastAPI Official Documentation – <https://fastapi.tiangolo.com>
- OpenAI API Documentation – <https://platform.openai.com>
- OpenCV Documentation – <https://opencv.org>
- WebRTC Documentation – <https://webrtc.org>
- Jurafsky & Martin – *Speech and Language Processing* (Book Reference)